IoT-Enabled Distribution Transformer Health Monitoring and Online IEEE C57.91 Thermal Insulation Degradation Estimation System

Abstract

Distribution transformers form the backbone of electrical power transmission and distribution infrastructure. However, they are continuously subjected to severe electrical, thermal, and mechanical stresses during operation. Unmonitored over-current conditions, oil leakages, and localized overheating accelerate insulation breakdown, often culminating in catastrophic equipment failure, prolonged power outages, and substantial capital loss. Traditional maintenance regimes rely heavily on periodic manual inspection or reactive protection devices that trip only after critical damage has occurred. Furthermore, existing digital monitoring setups focus solely on static threshold breaches without quantifying cumulative thermal degradation.

To address these limitations, this project proposes an intelligent, edge-computed IoT transformer health monitoring and predictive asset management system built around an ESP32 microcontroller architecture. The system establishes continuous, real-time sensing of critical operational parameters using low-cost, high-precision sensors:

- **ZMPT101B Module** for root-mean-square (RMS) AC voltage measurement,
- **ACS712 Module** for RMS load current tracking,
- **DS18B20 Waterproof Probe** for top-oil thermal telemetry, and
- **HC-SR04 Ultrasonic Sensor** for real-time oil reservoir level monitoring.

Unlike conventional telemetry nodes that merely log raw sensor values, the core novelty of this proposed system lies in embedding the **IEEE C57.91 thermal aging model** directly into the ESP32 edge firmware. The system dynamically computes the winding hot-spot temperature (Θ_H) using measured top-oil temperature (Θ_{top}) and load ratio:

$$\Theta_H = \Theta_{top} + \Delta\Theta_{H,rated} \times \left(\frac{I_{rms}}{I_{rated}}\right)^2$$

Using Θ_H, the firmware continuously evaluates the Arrhenius-based Aging Acceleration Factor (F_{AA}):

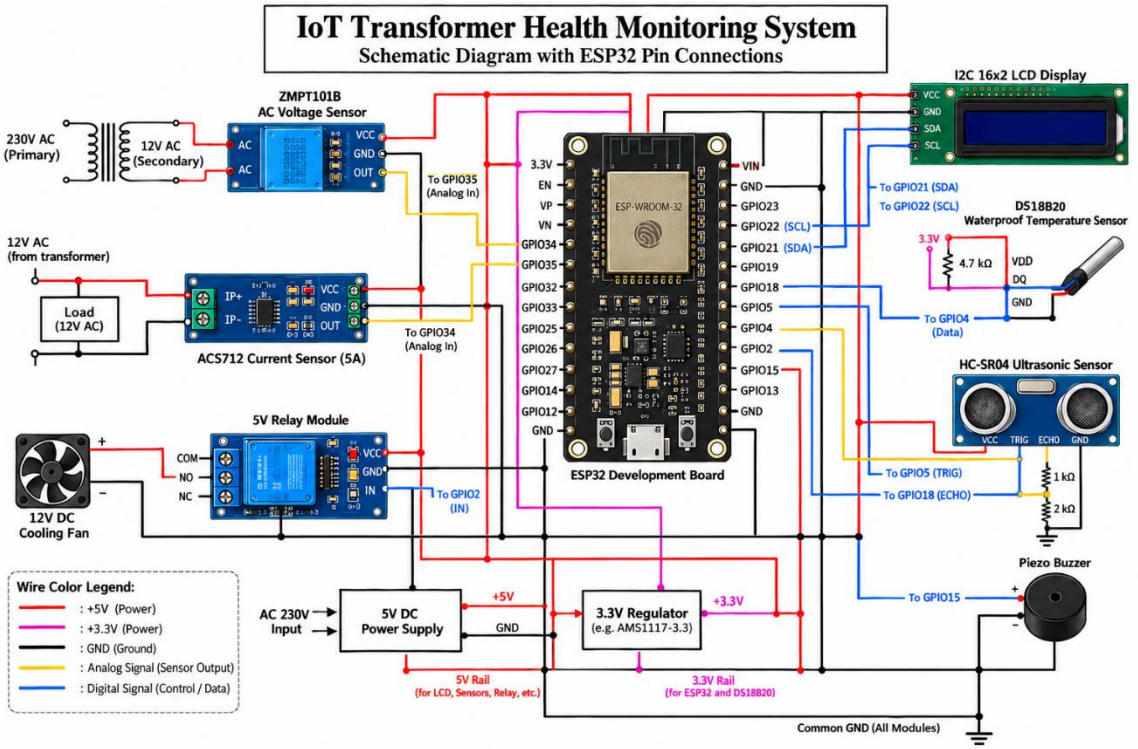
$$F_{AA} = \exp\left(\frac{15000}{383} - \frac{15000}{\Theta_H + 273.15}\right)$$

By integrating F_{AA} over time, the system computes cumulative Insulation Loss of Life (L_{loss}) and calculates the Remaining Useful Life (RUL %) of the transformer insulation in real time. Furthermore, an onboard edge diagnostic engine evaluates

multi-sensor states simultaneously to classify specific anomalies—such as severe overloads, cooling breakdowns, and oil leakage hazards—rather than issuing generic alarm signals. Local visual feedback is delivered via an I2C 16x2 LCD display, while remote monitoring, historical logging, and emergency alerts are transmitted wirelessly to the Blynk IoT cloud platform.

Proposed model

Proposed Outcomes of the Project



1. **Continuous Multi-Parameter Telemetry:** Simultaneous real-time acquisition and display of AC voltage, load current, top-oil temperature, and oil level on both local hardware (I2C LCD) and cloud interfaces (Blynk dashboard).
2. **Onboard Loss-of-Life (LOL) Estimation:** Operationalization of the IEEE C57.91 model on an embedded microchip, providing continuous live calculation of insulation degradation (F_{AA}) and Remaining Useful Life (RUL %) without requiring cloud-side post-processing.
3. **Edge Anomaly Discrimination:** Automatic root-cause identification distinguishing between electrical short circuits/overloads, cooling system fan failures, and physical oil leakage hazards.
4. **Automated Protection & Active Cooling:** Instantaneous triggering of a 5V relay module for forced cooling fan actuation during thermal overstress, alongside piezo buzzer activation for critical electrical faults.

5. **Low-Cost Prototyping:** A fully scalable, industrial-grade predictive asset management system demonstrated on a low-cost micro-budget hardware architecture.